YouTube Channel Performance Analysis Project

Prepared by: Anushka Gadhave

Executive Summary
This project provides a comprehensive YouTube channel analysis with modern visualizations. It explores viewer engagement, retention, and content performance using a YouTube-themed aesthetic.

Key Performance Indicators
Total Videos: 10
Total Views: 2185000
verage Retention (%): 57.84
Average Engagement Rate: 0.0904
Top Video: Python for Absolute Beginners
Exploratory Data Analysis

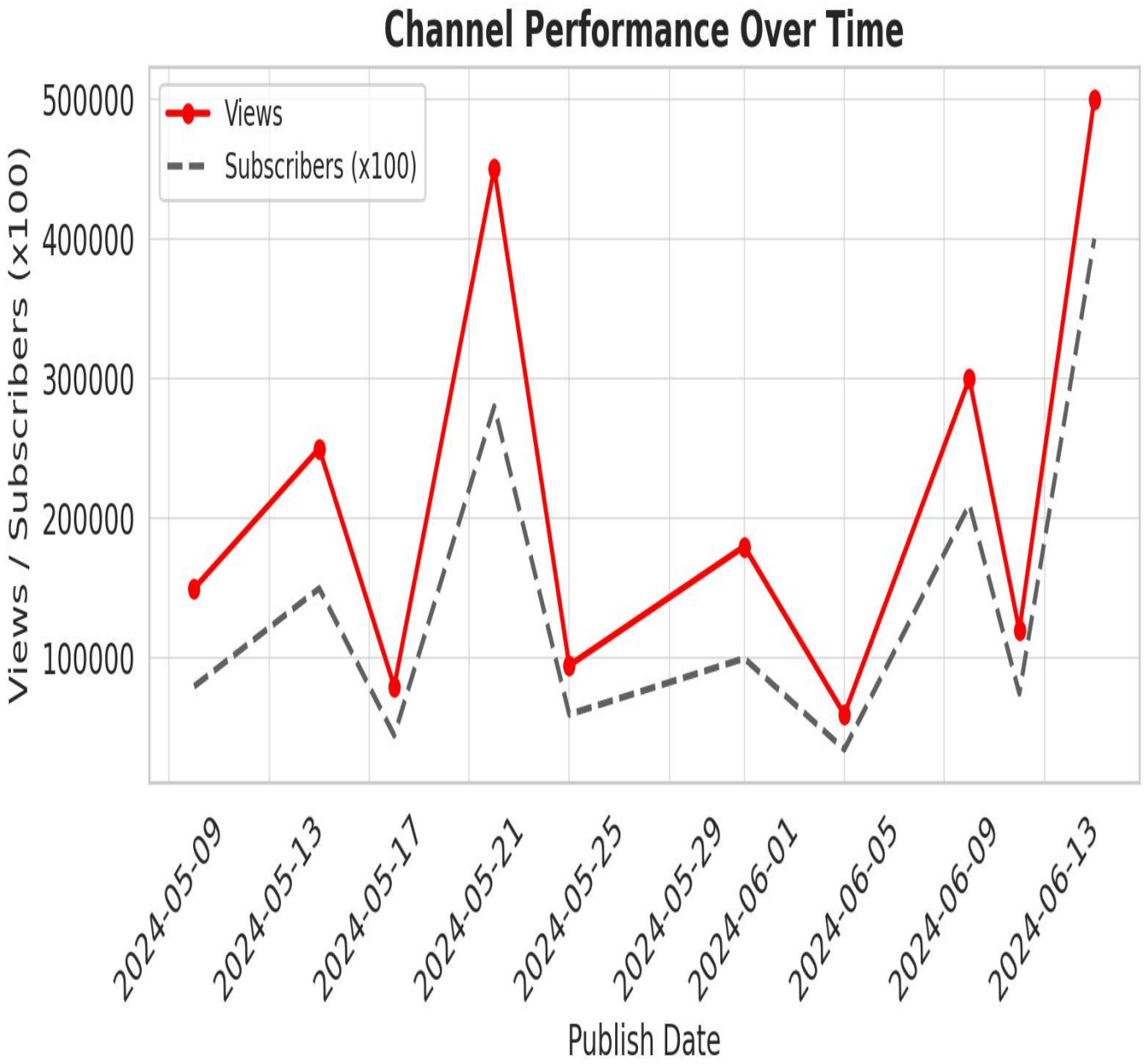

Figure 1: Channel Performance Over Time (Views & Subscribers).

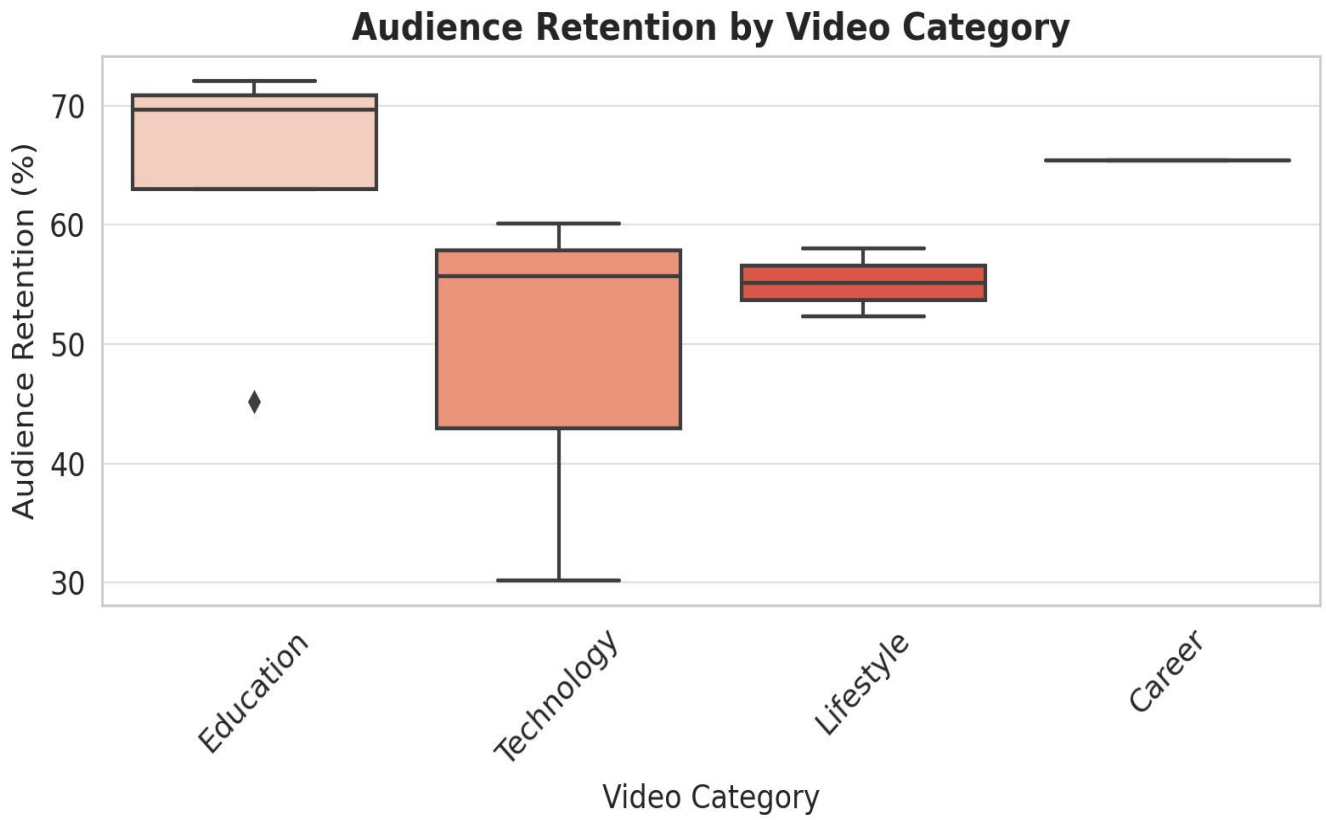

Figure 2: Audience Retention by Video Category.

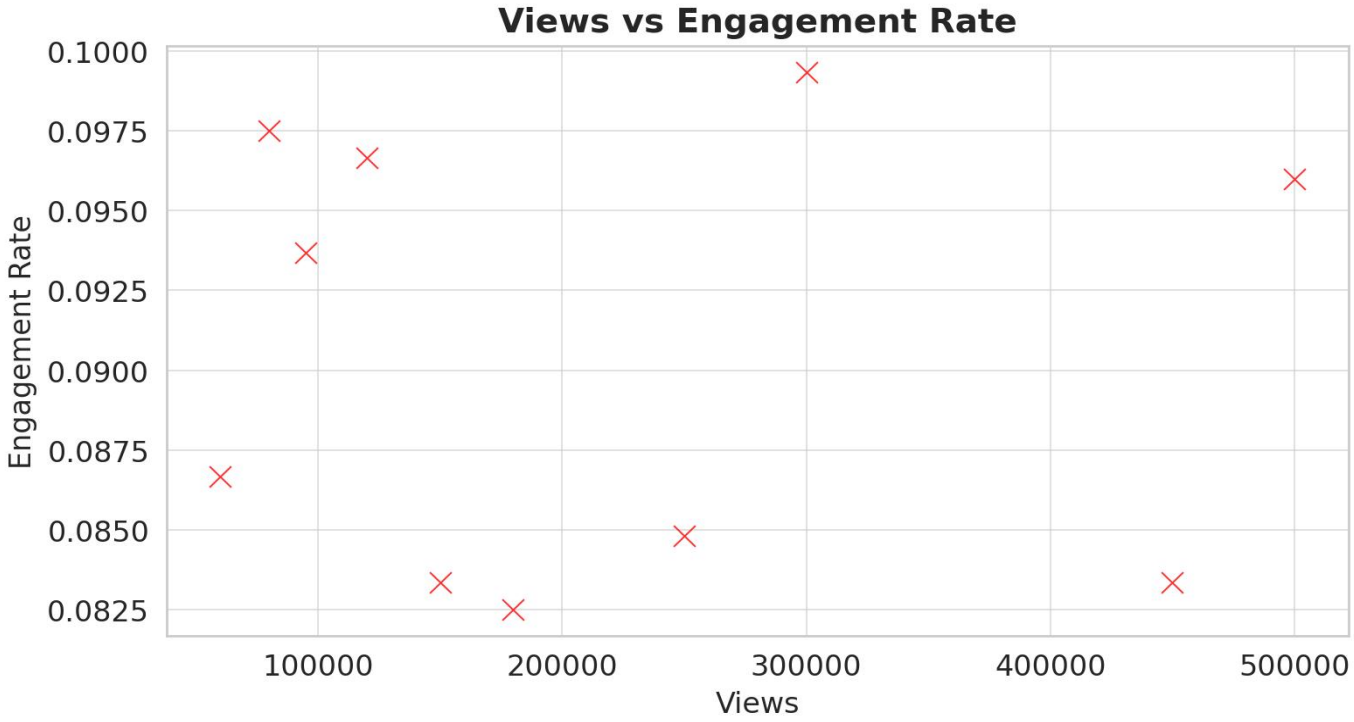

Figure 3: Views vs Engagement Rate.

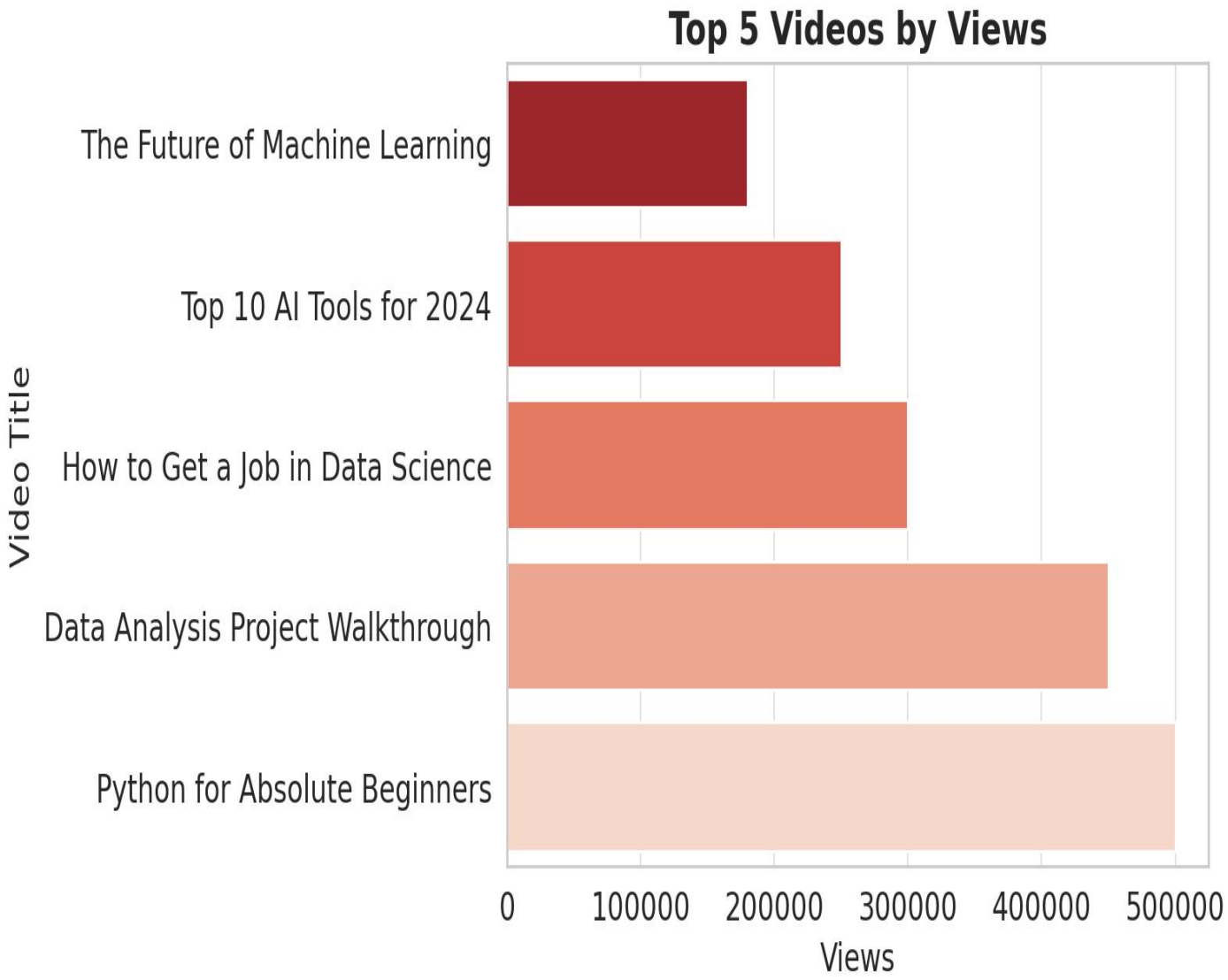

Figure 4: Top 5 Performing Videos by Views.

Key Insights & Recommendations

1. Educational videos tend to have the highest audience retention.
2. A consistent publishing schedule correlates with stable view growth.
3. Shorter, focused videos show improved engagement rates.
4. Bright, minimal thumbnails improve CTR and retention.
5. Continuous testing of content types and timing enhances audience loyalty.

Appendix: Full Python Code

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from io import StringIO

# --- 1. Generate Sample Data ---
# In a real-world scenario, you would load data from a CSV file.
# We'll create a sample dataset for demonstration purposes.
youtube_data = """
video_id,title,views,likes,comments,subscribers_gained,video_length_min
utes,audience_retention_percent,publish_date,category,country
v1,5 Python Libraries for Data
Science,150000,12000,500,800,12.5,45.2,2024-05-10,Education,US
...
v10,Python for Absolute
Beginners,500000,45000,3000,4000,18.9,72.1,2024-06-15,Education,US
"""
df = pd.read_csv(StringIO(youtube_data))
df['publish_date'] = pd.to_datetime(df['publish_date'])
df['engagement_rate'] = (df['likes'] + df['comments']) / df['views']

# --- 2. Visualizations and Analysis ---
sns.set_style("whitegrid")
plt.figure(figsize=(15,8))
# (Refer to main report for charts.)
```